Ishaan Patel

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SKILLS

- **Languages:** Python, C++, Java, C#, C, JavaScript, SQL
- **ML & AI:** PyTorch, OpenCV, Transformers, NumPy, Pandas, Scikit-learn, Optuna, LangGraph
- **Cloud & DevOps:** AWS, Azure, Docker, Kubernetes, PostgreSQL, Linux
- **Frameworks & Tools:** FastAPI, FastMCP, .NET, React, Spring Boot, Git

EXPERIENCE

PROMETHEUS GROUP • Software Engineer Aug 2024 - Jul 2025

- Delivered 15+ features and resolved 40+ defects across the flagship product in C# .NET and React; refactored legacy components using Domain-Driven Design and planned sprint work for new customer contracts.
- Architected the flagship product's automated testing framework as the first implementation of a company-wide QA automation initiative post-acquisition; running 100+ API, 30+ UI tests on every pull request and nightly smoke tests via GitLab CI/CD.
- Deployed production hotfixes triggered by automation alerts, reducing sprint regression testing time by 40%.
- Mentored incoming developers through paired programming and code review; the framework became the standard for automated testing, adopted by 3+ product teams.

FIDELITY INVESTMENTS • Software Engineer Intern Jun 2023 - Aug 2023

- Developed a Java Spring Boot service for 401(k) transaction processing using a layered DAO/service architecture with ORM persistence patterns; configured Jenkins CI/CD and migrated the codebase from Bitbucket to GitHub.
- Built a message routing pipeline on Azure Service Bus with dead-letter queue workflows to capture and retry failed messages.

UNIVERSITY OF NORTH CAROLINA CHARLOTTE • Undergraduate Researcher Jan 2023 - Jan 2024

- Adapted Microsoft's ALEX learned index to dynamic sparse graph workloads in C++, encoding edges as bitwise-concatenated 64-bit keys; benchmarked insertion and traversal performance against STX B+ tree and graph-optimized formats (CSR, VCSR, PCSR, BAL) across 10 real-world SNAP graphs on the UNCC HPC cluster.
- Parallelized PageRank, SSSP, BFS, and 1-hop workloads with OpenMP across 1–32 threads, measuring up to 20x speedup.
- Demonstrated 1.5–2.8x faster edge insertion with graph-encoded ALEX than the B+ tree baseline and identified why graph-specific formats outperform learned indexes on traversal.

FIDELITY INVESTMENTS • Software Engineer Intern Jun 2022 - Aug 2022

- Built REST APIs and Java AWS Lambda functions for the Fidelity Bloom debit card roundup feature, validated with JUnit and CloudWatch.
- Deployed AWS CloudFormation stacks (S3, SQS, Lambda, KMS) to dev/QA environments through Jenkins pipelines.

PROJECTS

Upwind - Real-Time Air Quality Intelligence getupwind.me | github.com/Ishaan1402/upwind

- Built and deployed a live full-stack app that informs users why local air quality spikes; processes AirNow, OpenAQ, NASA FIRMS, Open-Meteo, and satellite aerosol feeds to score the most likely causes and generate the explanation in plain English.
- Rule-based scoring ranks 5 common causes; an LLM judge verifies every explanation against the evidence and replaces failures with a safe fallback, so users only see honest explanations, logging each verdict to build a labeled dataset of failure cases from live traffic.

Pathfinder - Decoupled HPO Supervisor for Deep Learning github.com/Ishaan1402/pathfinder

- Built a hyperparameter optimization service with < 20 ms suggestion latency, early stopping, and monitoring for stalled or OOM runs. Uses SQLite WAL for concurrent reads and writes.
- Added a read-only MCP server enabling IDE agents to monitor live training convergence using trial telemetry and fANOVA importances, and to draft new HPO studies from existing training scripts pending manual review before registration.

crack-seg - UAV Bridge Crack Segmentation github.com/Ishaan1402/crack-seg | hf.co/ishaan1402/crack-seg

- Trained a 7.87M-param U-Net for pixel-level UAV bridge crack segmentation, achieving 0.730 Dice and 0.575 IoU on a held-out 447-image multi-source test set, tuned with Pathfinder.
- Deployed a FastAPI + Docker service to Hugging Face that handles 4K/8K UAV imagery via 448-px sliding-window inference with Gaussian blending; presented at the 2026 NCSU AI Student Symposium.

EDUCATION

NORTH CAROLINA STATE UNIVERSITY • M.S. in Computer Science, ML/AI Concentration Aug 2025 - May 2027
GPA: 3.78

UNIVERSITY OF NORTH CAROLINA AT CHARLOTTE • B.S. Computer Science Aug 2020 - May 2024
GPA: 3.85 | Magna Cum Laude | Chancellor's List